



AGENTIC AI FOR FINANCIAL SERVICES INDUSTRY

From Intelligent Assistance to
Autonomous Action





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Agentic AI: Executive Summary and Market Landscape

MARKET SIZE:

Global Agentic AI market expected to grow from \$5.1 billion in 2024 to **\$47.1 billion by 2030** (CAGR of 44.8%)

REVENUE:

Agentic AI can drive **3–15%** revenue uplift per relationship manager in banking.

COST SAVING:

30–50% cost reductions driven by automation of repetitive tasks, reduced manual intervention, and streamlined operational workflows

TIME SAVING:

Boosts autonomous decision-making efficiency by approximately **35%** and can reduce task execution time by up to **86%**

ADOPTION:

Approximately **45%** of Fortune 500 firms are piloting Agentic AI, with **USD 9.7 billion** in venture capital funding since 2023

PRODUCTIVITY POTENTIAL:

200% to 2,000% productivity gains in high-impact workflows like KYC checks, transaction monitoring, and fraud case reviews

Executive Summary

Financial services are entering a new phase of AI adoption. While Generative AI has been widely adopted, most deployments remain assistive and horizontal in nature. As a result, despite high adoption rates, GenAI initiatives have delivered limited measurable ROI and have largely left core financial workflows unchanged.

Agentic AI represents a fundamental shift in how AI is applied. Rather than merely supporting human decision-making, Agentic AI systems are designed to take ownership of defined, end-to-end processes. This shift creates an opportunity to re-architect complex, highly regulated financial workflows with greater speed, accuracy, consistency, and measurable business outcomes.

Real value from Agentic AI depends on deliberate strategic choices: where autonomy is applied, how responsibilities are allocated between humans and machines, and how risk, control, and accountability are governed.

To move beyond experimentation, organisations should treat Agentic AI as a process transformation capability, not a productivity enhancement tool. Deployments should focus on outcome-driven, high-impact workflows, supported by end-to-end process redesign and early integration of governance and control mechanisms. A buy-led technology strategy, while retaining ownership of decision logic and governance can help institutions realise durable and measurable business value.

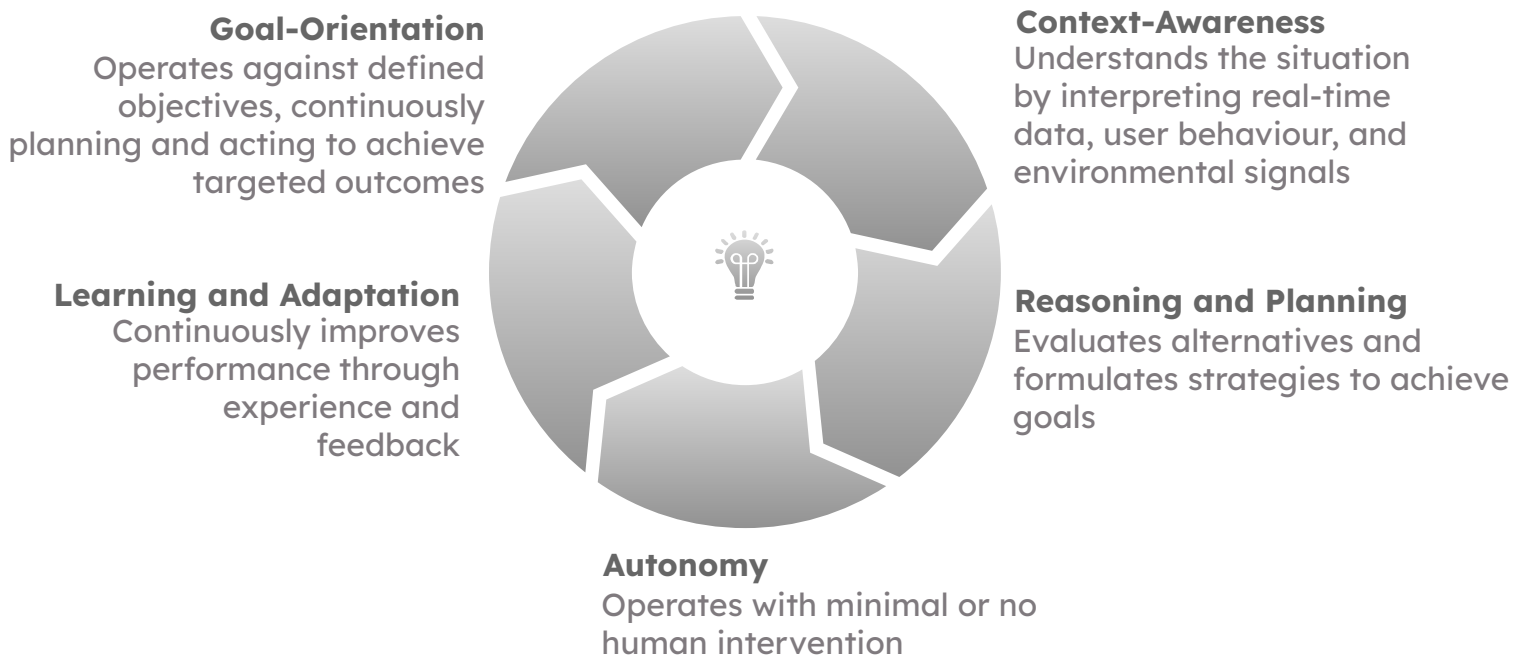
This report examines where Agentic AI delivers the highest business impact in financial services, the architectural capabilities required to deploy it safely at scale, and the emerging ecosystem enabling this transition, equipping decision-makers to move from experimentation to sustainable value creation.

What is Agentic AI?

Agentic AI refers to autonomous, goal-driven AI systems that plan and execute multi-step tasks with minimal human intervention, independently managing complex workflows such as loan approvals, fraud detection, and portfolio management. This represents a shift from task-level automation to goal-oriented intelligent autonomy, enabling banks and fintechs to operate with greater speed, precision, and personalisation

Agentic AI achieves autonomy by integrating Large Language Models (LLMs) with reasoning and planning engines, memory modules, and tool-use frameworks. These components work together through an orchestration layer that enables the system to interpret goals, evaluate options, and execute actions across enterprise systems without continuous human supervision.

Key Principles of Agentic AI



Evolution of Agentic AI

Timeline	Phase	Description
1940s–1970s	Foundational Ideas & Rule-Based AI	Early ideas of cybernetics, neural networks, and machine intelligence emerged. AI systems were rule-based and operated on predefined logic without learning or adaptation.
1980s–2000s	Machine Learning Era	AI shifted from purely rule-based systems to learning patterns from data using statistical and probabilistic methods, laying the groundwork for later advances.
1989–2017	Deep Learning & Reinforcement Learning Era	Multi-layer neural networks emerged around 1989 and became practically viable in the 2010s. Combined with reinforcement learning, AI learned to understand images, speech, and text and to optimise actions through trial-and-error.
2018	Large Language Models (LLMs)	AI gained strong language understanding and reasoning, forming the cognitive base for agents.
Early 2020s	Prompt Engineering	Techniques like chain-of-thought revealed step-by-step reasoning and task decomposition, showing early agent-like behaviour.
2022–2024	Emergence of Agent Framework Systems	AI systems began using tools, APIs, and external data to take actions. Multimodal models enabled understanding across text, image, audio, and data. Agent frameworks and emerging protocols enabled planning, memory, orchestration, and early agent-to-agent communication.
2025–2026	Autonomous Multi-Agent Systems Era	AI systems operate as autonomous agents capable of multi-step execution, with coordinated multi-agent collaboration and workflow integration across enterprise environments.

How does Agentic AI differ from GenAI?

Gen AI		Agentic AI
Generates content (text, code, insights) from data and prompts; enhances creativity and decision support.	Core Function	Acts autonomously toward defined goals by perceiving, planning, and executing multi-step workflows with minimal human input.
Reactive - responds to user prompts or predefined instructions.	Nature of Intelligence	Proactive - identifies objectives, plans actions, and executes tasks autonomously.
Focused on task-level or step-wise execution, such as drafting responses or content.	Workflow Capability	Manages end-to-end workflows by breaking them into subtasks and coordinating across multiple systems or tools.
Provides recommendation; humans retain responsibility for decisions and actions.	Decision-Making	Makes and executes decisions within defined controls, adapting to real-time conditions.
Learns from historical data but typically requires retraining for new contexts.	Adaptability	Continuously learns through interaction and feedback, adapting behaviour in real time.
Drafting risk summaries, generating customer responses, assisting analysts with reports.	Examples	End-to-end KYC/AML processing, credit assessment execution, proactive fraud monitoring, autonomous customer service actions.

Key Pain Points and Opportunities in Financial Services Industry

Inefficiency and Low ROI in KYC/AML Operations

Despite financial institutions in advanced markets increasing annual spending on KYC and AML compliance by up to 10% between 2015 and 2022, the industry continues to detect only approximately 2% of global financial crime flows.

Slow Decision Cycles in Financial Processes

Loan approvals, fraud detection, and compliance checks in traditional banking are slow and manual, often taking days or even weeks to complete, reducing overall operational efficiency.

Lack of Personalisation and Trust

Financial institutions struggle to deliver trusted, personalised experiences at scale due to siloed data, reactive AI models, and inconsistent decision logic. This limits their ability to provide contextual, explainable, and consistent financial guidance.

Legacy Infrastructure & Fragmented Data Architecture

Financial institutions often operate on fragmented, legacy infrastructure, and siloed data environments, which slows down underwriting, onboarding, risk decisions making and general responsiveness.

Regulatory Change & Policy Implementation Complexity

Translating new regulations into operational controls remains slow, manual, and error-prone, increasing both compliance risk and operational burden.

Error-Prone Risk and Compliance Management

Traditional risk and compliance systems remain highly dependent on manual review and fragmented rule sets, making them prone to human error in areas such as credit risk assessment, fraud detection, and regulatory compliance.

Consumers Lack Proactive Financial Optimisation

Most financial services require customers to actively manage their cash flows, debt, and investments, placing the burden of optimisation on the end user. As a result, customers frequently miss better financial options, optimal timing, and personalised recommendations.



Agentic AI Use Cases and Impact

Use Case	GenAI Does This	Agentic AI Additionally Does This	Concrete Example (GenAI → Agentic AI)	Real-World Impact (Directional)
Personalised Financial Planning	Provides on-demand, personalised insights on savings, spending, and investments.	Continuously monitors real-time financial data and autonomously rebalances portfolios, adjusts budgets, and executes decisions.	GenAI: Recommends portfolio rebalancing during market volatility Agentic AI: Automatically rebalances holdings, pauses SIPs when cash flow tightens, reallocates tax-saving instruments, and notifies the customer only if intervention is needed.	• 20–40% improvement in productivity / outcomes in AI-enabled frontline & advisory roles
Regulatory Compliance Automation	Summarises new regulations and assists with policy documentation.	Interprets regulations, maps them to controls, monitors transactions, and triggers remediation workflows.	GenAI: Summarises a new AML (Anti-Money Laundering) circular. Agentic AI: Updates compliance rules, adjusts monitoring thresholds, initiates internal audits, assigns actions to teams, and tracks closure automatically.	• 30–50% reduction in manual compliance effort
Autonomous Fraud Detection & Security	Detects potential fraud based on known historical patterns.	Learns new fraud behaviours, adapts rules dynamically, and coordinates real-time containment actions.	GenAI: Flags a suspicious transaction for manual review. Agentic AI: Detects a new scam pattern, tightens control in real time, temporarily restricts accounts, triggers customer verification, and escalates only confirmed cases.	• 25–60% reduction in false positives
KYC/AML (Anti-Money Laundering)	Extracts documents, screens names, and summarises transactions data.	Executes the full KYC/AML lifecycle end-to-end, escalating only true exceptions to humans.	GenAI: Extracts PAN/ Aadhaar details and summarises customer profile. Agentic AI: Collects documents, verifies identity, screens sanctions, monitors post-onboarding behaviour, drafts STRs, and escalates high-risk cases only.	• Onboarding reduced from days to minutes • 2–20x Improvement in Operational Productivity

Use Case	GenAI Does This	Agentic AI Additionally Does This	Concrete Example (GenAI → Agentic AI)	Real-World Impact (Directional)
Credit Risk Memo Creation	Drafts sections of a credit risk assessment memo using user provided data.	Autonomously gathers data, identifies gaps, runs analysis, and delivers a complete credit risk assessment memo.	GenAI: Drafts a financial summary for a loan proposal. Agentic AI: Retrieves bureau + GST + bank statements, identifies missing data, request clarification, runs risk models, and delivers a ready-to-use credit risk memo.	• 20 to 60% increase in productivity • 30% improvement in credit turnaround
Legacy System Modernisation	Assist developers with code generation and refactoring on request.	Coordinates multi-agent execution across refactoring, testing, validation, and system integration.	GenAI: Refactors a COBOL module when prompted Agentic AI: Plans modernisation, rewrites services, runs regression tests, validates integrations, flags risks, and reports progress to architects.	• >50% reduction in delivery time

Agentic AI Architecture Layers

1. Intelligence

Foundation Models / Reasoning Engines

- Provides the core cognitive capabilities of the system, including understanding language, reasoning, planning, and generating decisions.
- Rather than merely answering questions, agentic architectures reason about goals, trade-offs, and next actions. They rely on models capable of multi-step planning and decision-making, not just content generation

2. Store & Recall Context

Memory, Knowledge & Retrieval

- Enables the system to retain context, remember past actions, and learn from outcomes over time.
- Moves beyond “stateless responses” to continuous behaviour, allowing the system to learn from past actions, avoid repeating mistakes, and build institutional memory.

3. Build & Deploy Agents

AI Agent Provider

- Delivers pre-built or configurable AI agents that execute end-to-end business tasks, such as KYC processing, fraud investigation, and credit assessment within defined operational and governance boundaries

4. Plan & Coordinate

Orchestration

- Controls how tasks are sequenced and coordinated across one or multiple agents.
- Determines execution flow, manages dependencies, routes work between systems, and enables human-in-the-loop approvals when required.

5. Execute Actions

Tooling & Action

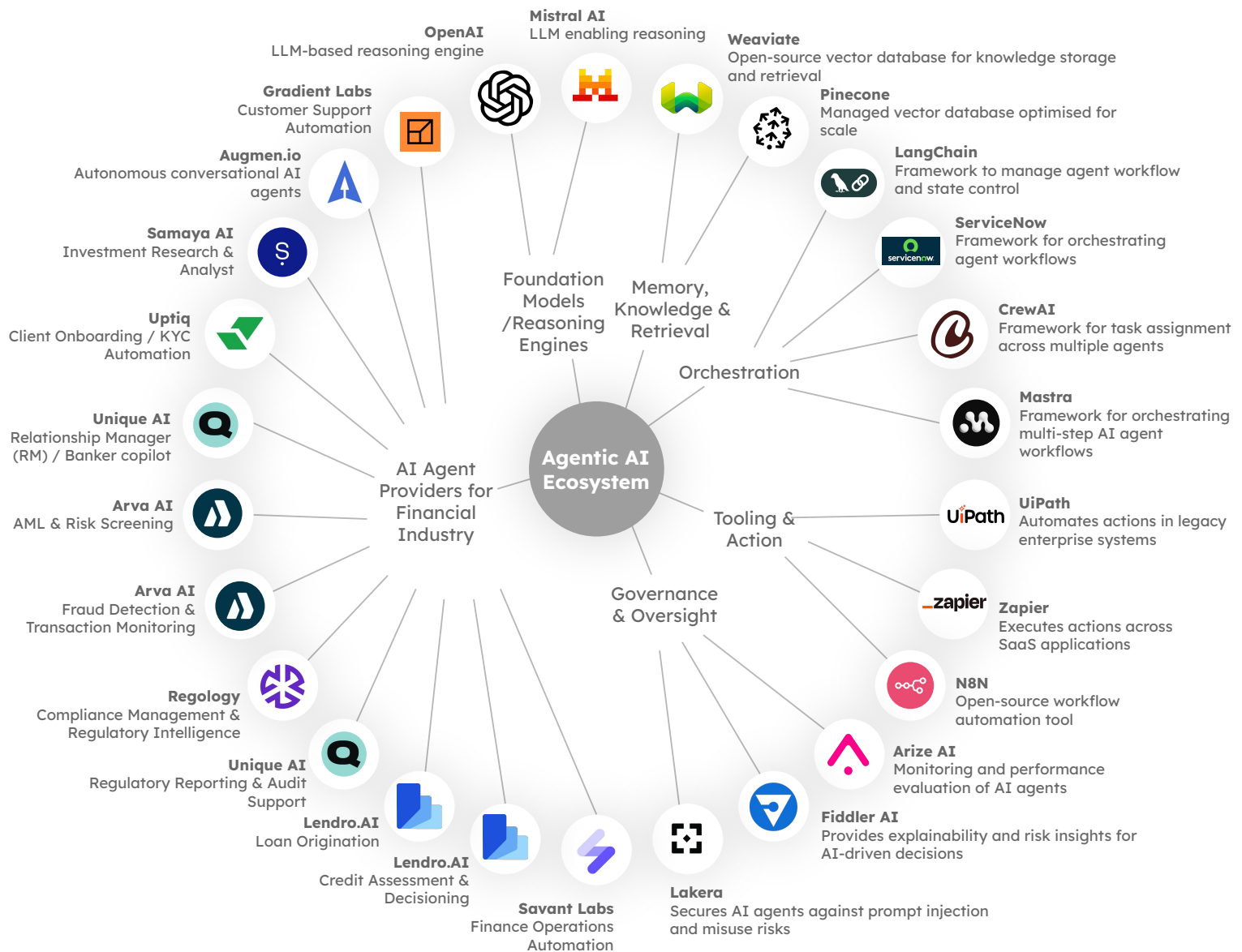
- Allows AI agents to take actions in real business systems through APIs, RPA, and applications integration..
- Connects AI decisions to execution by updating data, triggering workflows, and completing tasks automatically.

6. Control & Govern

Governance & Oversight

- Ensures that the AI systems are monitored, auditable, safe, and controllable.
- Strong agentic AI includes accountability, audit trails, approval flows, and “kill switches” enabling trust at scale. Robust governance differentiates deployable enterprise-grade systems from experimental demos.

Agentic AI Ecosystem



Commonwealth Bank of Australia

Project Coral: Enhancing the Engineering Experience with Agentic AI

Project Coral is an internal Agentic AI framework developed by the AI Powered Engineering team at Commonwealth Bank of Australia (CBA). The platform uses multiple AI agents to automatically identify technical issues across the software development lifecycle and propose fixes, improving code quality, security, and engineering velocity.

Aim: To proactively eliminate technical debt, reduce repetitive work, and improve code quality, security, and engineering focus

Challenges Addressed:

- 1) **Disparate tooling and fragmented visibility** - Engineers were required to manually monitor multiple tools to identify issues related to code quality, security, and performance.
- 2) **Lack of coordinated automation** - Existing AI tools or assistants operated in isolation and required manual intervention, limiting their ability to deliver end-to-end workflow automation.

Solutions:

- As part of its AI-Powered Engineering programme, CBA launched Project Coral, an Agentic AI framework supporting 7,800+ engineers across the organisation.
- The framework deploys multiple specialised AI agents, including:
 - 1.Code Fix Agent – Identifies and resolves code quality issues
 - 2.Test Agent – Ensures adequate test coverage
 - 3.Security Agent – Detects security vulnerabilities
 - 4.PR Review Agent – Validates coding standards and best practices
- These agents autonomously detect technical debt (e.g., performance slowdowns, legacy issues), analyse code repositories, and generate pull requests with recommended fixes. Human engineers review and approve the changes before deployment.

Impact:

- Improved early detection of technical debt, security vulnerabilities, and code quality issues.
- Over 70% of engineers now use AI-powered tools daily.
- Achieved a 30% increase in code changes, reflecting higher engineering productivity and faster delivery.

Capital One: Agentic AI Enhancing the Car Buying Experience

Capital One is a U.S.-based financial services company offering banking, credit cards, and lending products, including auto finance.

To strengthen its auto financing business while delivering a seamless car-buying experience, Capital One developed Chat Concierge, a multi-agent conversational AI system that guides customers through the car-shopping journey while enabling dealers to engage online buyers more effectively. By engaging customers early, before financing decisions are finalised, Capital One improves the overall digital experience and increases the likelihood of being selected as the preferred auto-loan provider.

Challenges Addressed:

Car buyers increasingly expect transparency, convenience, and digital guidance when making high-value purchases. At the same time, auto dealers often struggle to consistently engage online shoppers and convert digital interest into qualified leads. Financial institutions typically enter the purchase journey late, limiting influence over customer decisions and increasing acquisition costs.

Solutions: Chat Concierge

- Capital One developed Chat Concierge, an internally built, multi-agent conversational AI system designed to support customers throughout the car-shopping journey.
- Powered by agentic AI, Chat Concierge goes beyond answering questions and can autonomously execute multi-step actions, including: Comparing vehicles and features, Estimating affordability and Finance options, Coordinating next steps like test-drive scheduling.
- At the system level, multiple specialised agents collaborate: One interacts with the customer, another determines the next best steps, a third enforces rules and compliance, and a fourth clearly communicates the outcome. This coordinated goal-driven approach enables customers to navigate complex, high-value purchase decisions more effectively than a traditional chatbot.

Impact:

- 24/7 customer engagement through automated conversational interactions
- Up to 55% improvement in engagement metrics
- Faster response times, reducing early-stage customer drop-off
- Lower operational effort through automated, multi-step interactions

Key Risks and Governance for Agentic AI

- As Agentic AI systems operate with greater autonomy across interconnected workflows, they introduce new system-level, data, and decision making risks. Strong governance frameworks are therefore essential to ensure operational control, clear accountability, transparency, and ongoing regulatory alignment.
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Risk Category	Risk Description	Governance Control
Chained System Vulnerability	Failures or compromises in a single agent or tool can cascade across interconnected workflows, amplifying impact at the system level.	Defined Autonomy Boundaries & Human Checkpoints
Authority & Access Overreach	Agents may exceed intended decision authority or system access permissions while interacting with enterprise platforms, tools, or APIs.	Identity, Access & Permission Control
Data Integrity & Information Exposure	Autonomous agents may expose or transfer sensitive customer and transactional data in ways that are difficult to detect or monitor in real time.	Data Governance, Monitoring & Auditability
Regulatory Misalignment	Autonomous actions may breach regulatory or internal policy requirements if compliance constraints are not embedded and continuously updated within the system.	Continuous Policy Enforcement & Review
Decision Transparency & Auditability Risk	Agent decisions may lack sufficient transparency, policy alignment, or regulatory traceability, increasing compliance and audit risks.	Traceability & Explainability
Human Oversight & Control	Over-reliance on autonomous agents may reduce human oversight, allowing decision drift or misaligned objectives to persist unnoticed.	Human-in-the-Loop (HITL) for High-Risk Decisions

Key Takeaways

Prioritise High-Impact, End-to-End Workflows

Financial institutions should prioritise Agentic AI deployments in complete, outcome-driven workflows such as KYC/AML, credit assessment, fraud investigation, and regulatory compliance, areas where GenAI has delivered limited ROI. Value emerges when agents own entire workflows, reducing cycle times, operational costs, and human error while delivering measurable business outcomes.

Re-Architect Workflows Before Deploying Agents

Agentic AI cannot deliver full value if layered on top of fragmented legacy, processes. Institutions should re-architect priority workflows end-to-end, redesigning them around autonomous execution with clearly defined goals, exception handling, and human checkpoints. Treating Agentic AI as a process transformation lever rather than a productivity add-on can unlock step-change improvements in turnaround time, cost efficiency, and operational resilience.

Embed Governance, Control, and Accountability

Agentic AI adoption must be accompanied by strong governance frameworks from day one. Institutions should define clear autonomy boundaries, approval checkpoints, audit trails, and “kill switches,” to ensure regulatory alignment, transparency, and accountability. Human-in-the-loop mechanisms should remain mandatory for high-risk decisions to maintain trust and regulatory confidence.

Build vs Buy Approach

Financial institutions should buy foundational AI infrastructure for a competitive edge while building internal control layers for oversight. Driven by data security and regulatory compliance, organisations must maintain internal ownership of the data layer, decision workflows, governance rules, and audit mechanisms to ensure risk accountability and customer outcome ownership. Exploring solutions such as Small Language Models (SLMs) can help bridge the gap between innovation and safety, enabling a secure, scalable AI ecosystem aligned with Financial Services Standards.



We can enable your journey in this rapidly changing disruptive economy. OpenI Platform highlights:

- Next disruptor you must know about
- How your competitors are engaging with these disruptors/ innovators
- Startup ecosystem to drive your growth

Navigate with Confidence

BUSINESS GROWTH

Identify high potential startup investment opportunities

STRATEGY & INNOVATION

Source new business models, spot future competitors

BUSINESS OPERATIONS

Optimise business operations

MARKETING

Drive sales and customer engagement (D2C)

PARTNER MANAGEMENT

Procure and manage next-gen disruptors

ASK OUR ANALYSTS

White Space Analysis
(Spidermap)

Ecosystem as a
Strategy

Managed
Sourcing

Corporate
Venture

Competitive
Analysis

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